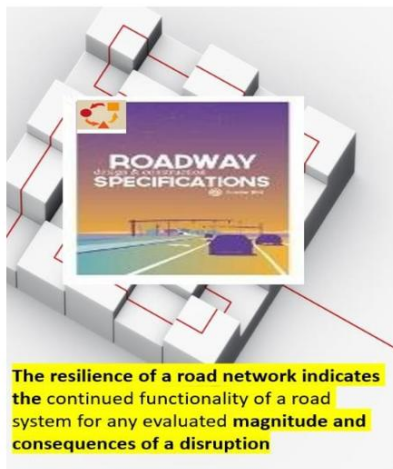


Possible must-do(s)

- ✦ A RS&A package must have a strategy to correlate V2R effectiveness impacting details
- ✦ A RS&A package must have a strategy to design V2R strategies and risk mitigation
- ✦ A RS&A package must have a strategy to conduct random sampling of V2R effectiveness or VSL information
- ✦ A RS&A package must have a strategy to conduct random sampling of V2R performance information like Driver Fitness, Vehicle Fitness, Road System understanding, Alpha Assistance contribution and Call to attention methodology improvement, so there is no issue of misinformation in reporting effectiveness, responsiveness and planned performance.

Perspective Imagery (PI) Indicators for a road system/route



To be completed

For the science or insight behind FESA or NSSR Theme based projects, it needs to be said that Sustainable Climate Change Mitigation will work using a geo-linked LAW or CYCLE of resilience



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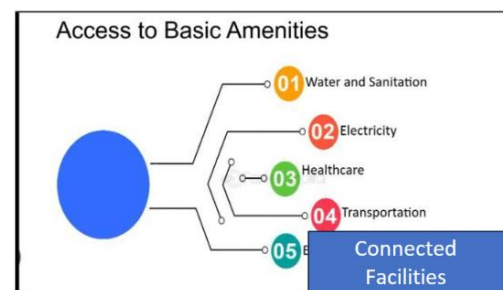
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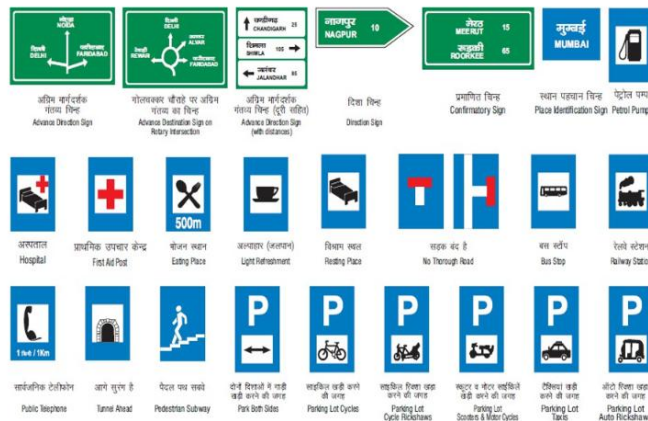
Civic Amenity RI-FESA NOC Culture



(FESA stands for Fire and Emergency Services Actuation)

Civic Amenities / RI FESA NOC Culture Links

• No objection Certificate (NOC) Culture to help NSSR RS/FESA



Valuable examples of Civic Amenities

- ☐ Traffic Control / Traffic Police Stations
- ☐ Pedestrian Subway/ Subway
- ☐ Railway Station
- ☐ Metro lanes / Tram track
- ☐ Bus stop
- ☐ First aid and Ambulance Post
- ☐ Hospital
- ☐ Connected Socio-Economic-Need and Incidence mitigation facilities
- ☐ Connected Vehicle Maintenance & Repair Facilities
- ☐ Connected Vehicle Spare parts suppliers
- ☐ Connected Cluster of suppliers
- ☐ Connected Power and Electricity management / Water supply/ Sanitation
- ☐ Connected Lean Waste management
- ☐ Place identification sign, Confirmatory sign
- ☐ Direction sign, Advance Direction sign

Design for FESA Performance

Focus	NSSR RS Leadership	Recommended Approach
Real world environment	↔	Field book for FESA NOC Culture, Trustworthy Facts and Assist FESA Analytics Help Desk system
Real world goals / targets	↔	FESA Performance Measurement Metrics/ KPI(s)
Real world strategies	↔	Process Knowledge and Compendium
Real world Tactics / Operations	↔	Assisting Analytics based insights for FESA-NOC Resilience and including a Deep Interaction link for Designs, Solutions, Projects, Services, Juran Trilogy Programmes
Real world Results	↔	Outcome of unified performance based initiative
Deliverables: Datasets, Spreadsheets, Reports, Dashboards, Field book, KPI handbook, FESA NOC Culture		

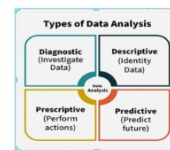
Nature of planning (Rated as a crucial influencer):

- () **Design standards compliance** (width of road, margins for pillars, gradient designs, curves designs, median designs, arboriculture safety, pedestrian and passenger safety, safe commuting between 2 points, reasonable time taken to travel from one point to another, enablers for vehicles that use renewable energy)
- () **Accountability for Traffic factors** (speed standards set for road systems, reaction time based on PIEV*, navigation standards, safe stopping sight distance, safe overtaking or passing, safe sight distance for entry into any associated intersections, feedback systems)
- () **Accountability for Environment factors** (sentinel screening and risk mitigation for unforeseen snow fall, hailstorms, heavy rainfall, thunder storm and lightning arrestors, ease of maintenance despite severe weather conditions)
- () **Maintenance Systems reliability** (proper design out maintenance, risk mitigation & maintenance, inspection and maintenance of extensions, gradient-design validation, policy for emergency services, policy for disaster management services)
- () **Quality of associated Drainage systems** (design and implementation after consideration of water table, sub-grade soil, reinforced earth, nature of geo-grids that are to be used in the road construction, management of seepage flow & capillary rise, reliable impervious wearing surface of road with aggregators and binders)
- () **Quality of traffic signalling systems** (“(Google Earth related) satellite imagery, or drone flight imagery or sentinel sensor feedback based” Risk Mitigation Desk notifications and proactive responses by the traffic management network, by nature of design “intelligent signaling solutions” that decide as to how traffic has to be managed or routed in case there is a disaster, accident, or in a case where part of the road or road system is rendered unusable)



Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

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- () **Satisfactory Emergency Response planning** (Equipped with signage and barricade deployment, contact numbers for nearest “ambulance services, hospital, police station, fire department, disaster management department”, availability of first aid provisions, equipped with fire extinguishers & fire fighting facilities, equipped with smoke alarm systems, equipped with sentinel sensors, has clearance for air lift to save life, has collapsible floor/ground escalation systems at designed locations to help evacuate passengers from elevated metro railways)

b. Nature of congestion (Rated as important negative influences):

- () Perennial congestion () Seasonal congestion
- () Time-based congestion
- () Incidence specific congestion
- () Feeder Traffic specific congestion
- () Goods/Freight movement specific congestion
- () Congestion due to other influences

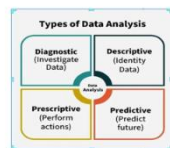
c. Stabilizing aspects (Rated as positive influences):

- () Has a Management Index Specification () Has satellite images
- () Included in Google maps () Is of good quality
- () Has multiple-lanes
- () Has sensor-enabled medians or bordering road barricades
- () Has reliable traffic signals
- () Has SMART Meters for immersive TEPO
- () Accountable traffic intervention possible at location
- () Not in close proximity to industries
- () Not in close proximity to rivers and other rainfall affected water bodies,
- () Has storm water drains
- () Has well maintained manholes and septic systems
- () Not affected by festivities
- () No pedestrian sidewalks
- () No encroachment



Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

ROAD SAFETY-OUR NATIONAL SAFETY AND SOCIAL RESPONSIBILITY





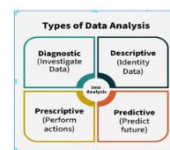
Strategic-Tactical-
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Management
influencers

- () No alteration
- () Not sidelined by trees
- () No afflicted by dumping of industrial waste
- () Not afflicted by dumping of public waste
- () Has a proper sewage system

d. Probable Hazards (Rated as very important negative influences):

- () Is an inter-link for other roads or routes etc
- () Is in close proximity to neighboring states
- () Is in probable or escalated tension areas
- () Is a sensitive area (where satellite imagery a threat)
- () Is in close proximity to an industrial cluster
- () With curving meanders
- () Has a steep incline with improper entry or exit
- () Has underlying dangerous landforms
- () Is in close proximity to dangerous landforms
- () Has a history of unattended potholes () Has potholes
- () Is sidelined by less maintained trees
- () Is in close proximity to rivers and other rainfall affected water bodies
- () Is in close proximity to marshes or swamps
- () Is part of a bridge or connects to a bridge
- () No storm water drains
- () Has poorly maintained manholes and septic systems
- () Afflicted by incidences of bottlenecks
- () Is difficult to manage via surveillance
- () **Is prone to crime** (due to lack of surveillance/being a remote location/ lack of traffic signals/lack of lighting)
- () **Is prone to accidents** (due to lack of sufficient planning for pedestrian and passenger safety)

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AND SOCIAL
RESPONSIBILITY



Strategic-Tactical-
Operational
intelligence
Infrastructure
and/or Data
Management
influencers

e. Associated planning, risk mitigation, repair and/or restoration programmes

The addressing of problems is either well-planned or not well-planned, where there are selective classifications that can help identify issue levels for the commuter:

Planned (Rated as positive influences)

- () Forecast based
- () Control Room based
- () In time surveillance based

Not well-planned (Rated as very important negative influence)

- () Only reciprocal (where problems are addressed in a reactive manner)
- () Only when problems are escalated
- () Only when mass grievances are reported

f. Signage deployed to mitigate risks to commuters or people

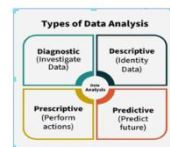
() **Road signs identifying traffic safety norms** (one-way or two-way signs, permitted timings, speed limits, rules for pedestrian and passenger safety, rules about overtaking, rules against cutting lanes, rules for parking, signage about low visibility zone, low height clearance and load levels)

() **Signage for accident relief, emergency response and assistance** (like contact information for the nearest "ambulance services, hospital, police station, fire department, disaster management department", associated civic body)

() **Signage and barricades around (perimeter) of potholes, poor quality manholes and septic systems**

() **Signage with precautionary and must know information about ring road, flyover, bridge, tunnel, subway, metro track, tram track, and level crossing**

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RESPONSIBILITY



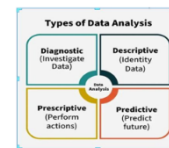


Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

Defect liability systems (Rated as a crucial influencer):

- () Road system/route utilization analytics / guidelines
- () Associated Heavy Rain/Storm/Disaster Warning analytics / guidelines
- () Associated Road system/route utilization related FESA / unplanned for driving conditions- based planning analytics / guidelines
- () Associated or critical NOC compliance/ commissioning status analytics / guidelines
- () Associated or critical NOC compliance/ commissioning status specific planning-evaluation-resolution schedules
- () Associated or critical Transformational solutions or Civic Amenities enabling

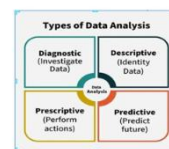
ROAD SAFETY-OUR NATIONAL SAFETY AND SOCIAL RESPONSIBILITY



Traffic Management strategy (Rated as a crucial influencer):

- () Current solutions for traffic management and road system/route utilization
- () OTA Theme based solutions for traffic management and road system/route utilization
- () Commuter subscribed -for solutions for traffic management and road system/route utilization
- () Building/Site/Plot owners/associations subscribed -for solutions for NOC adherence/NSSR Theme incorporation
- () Strategic-Tactical-Operational intelligence specific Infrastructure and/or Data Management

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Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers



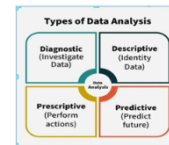
Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

ACCIDENT RELIEF, EMERGENCY RESPONSE AND ASSISTANCE **strategy (Rated as a crucial influencer):**

Docked View and Logistics for traffic management and road system/route utilization based on

- () FESA profile registration for buildings/road systems/roads
- () FESA scheduled activation for buildings/road systems/roads
- () FESA priority activation for buildings/road systems/roads
- () FESA Desk/Control Centre activation for buildings/road systems/roads
- () FESA 80:20 Rule of insightful analysis for buildings/road systems/roads

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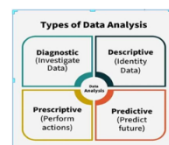
Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

Associated planning, risk mitigation, repair and/or restoration **strategy (Rated as a crucial influencer):**

Docked View and Logistics for traffic management and road system/route utilization based on

- () NOC Blocks of Assets/Liabilities/C2A Viewpoint data
- () NOC Profile data
- () FESA Scheduled data
- () FESA Incidence or Problem data
- () FESA Desk/Control Centre data in terms of messages/service related guidance/incidence or problem controlling guidelines
- () FESA 80:20 Rule of insightful analysis data

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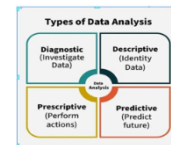
Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

Associated planning, risk mitigation, repair and/or restoration **strategy (Rated as a crucial influencer):**

Docked View and Logistics enabling for traffic management and road system/route utilization based on

- () Registration for NOC Blocks of Assets/Liabilities/C2A Viewpoint/Pilot recording into a Call to Attention FESA Hub
- () Subscription for NOC Blocks of Assets/Liabilities/C2A Viewpoint/Pilot recordings from a Call to Attention FESA Hub
- () Subscription for FESA/NSSR Theme Information from a Call to Attention FESA Hub
- () Subscription for FESA/NSSR Theme Process guidance from a Call to Attention FESA Hub
- () Subscription for FESA/NSSR Theme PER guidance from a Call to Attention FESA Hub, where PER stands for Planning, Evaluation and Resolution

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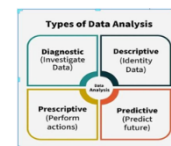
Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

Viewpoint Management for road system issues or incidence indicators **strategy (Rated as a crucial influencer):**

Docked View and Logistics enabling for traffic management and road system/route utilization based on

- ☐ FESA First View functionality
- ☐ FESA Service Anywhere Anytime drills or runs, operations and recommendation services
- ☐ Updated/connected (OTA enabled) call to attention imagery or perspective imagery

ROAD SAFETY - OUR NATIONAL SAFETY AND SOCIAL RESPONSIBILITY





Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

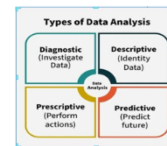
Viewpoint Management for road system issues or incidence indicators strategy (Rated as a crucial influencer):

Docked View and Logistics enabling for traffic management and road system/route utilization based on

❑ Updated/connected (OTA enabled) call to attention imagery such as Key Performance Indicators (KPIs) for a road system/route/NOC blocks of buildings with location relevance

❑ Updated/connected (OTA enabled) perspective imagery such as Perspective Imagery (PI) Indicators for a road system/route with location relevance and active deployment of Planned Road signs/Relatedly Live Screens/Billboards/Hoardings/Signages, where these views or insights can guide any FESA service or SAR experience, it is known that the BBMP has time and again banned road side hoardings to avoid these from being a distraction to drivers/commuters etc

ROAD SAFETY-OUR NATIONAL SAFETY AND SOCIAL RESPONSIBILITY



NOC related details for a building

KSFES - Portal Application

The Application for No Objection Certificate page provides the user with the following information:

- **Building Type as per NBC:** Select the building type as per NBC.
- **Building Usage Purpose:** Select the purpose of building usage.
- **Application for extension to an existing building or new proposal?:** Specifies whether the application is for extension to an existing building or new proposal.
 - **Extension to existing building:** Choose this option, if the application is for extension to an existing building.
 - **New Proposal:** Choose this option, if the application is for new proposal.
- **If Extension to an existing building, provide details for existing building:** If Extension to an existing building, enter the details as follows. This option will be available only if you have chosen the **Extension to existing building** option.
 - **Height (m):** Enter the height of the existing building in meters.
 - **No. of Floors:** Enter the number of floors
 - **Total Built up Area (sqm):** Enter the total built up area in square meter.
- **Whether Sanction obtained from Local Authority (BBMP,BDA etc) to construct the building?:** Choose the option which specifies whether the sanction is obtained from local authority to construct the building. This option will be available only if you have chosen the **Extension to existing building** option.

NOC Blocks of Buildings with Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

NOC related details for a building

KSFES - Portal Application

Volume 1

- **Site Details:** Here user can enter the details of the site/plot.
 - **Survey Number:** Enter the survey number of the plot.
 - **Survey Aksharabaga:** Enter the survey aksharabaga.
 - **Hissa No.:** Enter the Hissa number of the plot.
 - **Land Extent:** Enter the extended land use.
 - **Unit of Measurement:** Select the unit in which the plot is to be measured.
 - **Taluk:** Enter the taluk in which the project is proposed.
 - **Hobli:** Enter the hobli to which the plot belongs.
 - **Village:** Enter the name of the village.
 - **Ward Number:** Enter the number of the ward.
 - **Street:** Enter the name of the street.
 - **Pincode:** Enter the pincode of the city.
 - **PID (If Available):** Enter the PID code if available.
 - **Add:** Click on this button to add the site/plot details entered.
- **Location Coordinates of the site:** Here user can provide the latitude and longitude of the site.
 - **Latitude:** Enter the latitude of the site.
 - **Longitude:** Enter the longitude of the site.

NOC Blocks of Buildings with Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

NOC related details for a building

KSFES - Portal Application

- **Chakbandi:** Specifies the chakbandi details of the plot.
 - **North:** Enter the name of the person who owns the north side of the land.
 - **South:** Enter the name of the person who owns the south side of the land.
 - **West:** Enter the name of the person who owns the west side of the plot.
 - **East:** Enter the name of the person who owns the east side of the plot.
 - **Add:** Click on this button to add the chakbandi details entered.
- **Entry Road Type:** Select the type of entry road to the site.
- **Entry Road Name/Identification:** Enter the name of entry road.
- **Entry Road Width (m):** Enter the width of the entry road in meter.
- **Exit Road Type:** Select the type of exit road from the site.
- **Exit Road Name:** Enter the name of the exit road.
- **Exit Road Width:** Enter the width of the exit road in meter.

NOC Blocks of Buildings with Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

NOC related details for a building

KSFES - Portal Application

- **Setback as shown in Plan (m):** Here provide the setback details of the plot as shown in plan.
 - **Front:** Enter the setback area in front of the proposed building.
 - **Rear:** Enter the setback area in the rear side of the proposed building.
 - **Left:** Enter the setback area in the left side of the proposed building.
 - **Right:** Enter the setback area in the right side of the proposed building.
 - **No. of Floors:** Enter the number of floors in the proposed building.
 - **Height of the Building (m):** Enter the height of the building in meter.
 - **Total Built-Up Area (sqm):** Enter the total built up area of the building in square meter.
 - **Provided Parking for Two Wheelers:** Enter the number of two wheelers for which parking is provided.
 - **Provided Parking for Four Wheelers:** Enter the number of four wheelers for which parking is provided.

NOC Blocks of Buildings with Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

NOC related details for a building

KSFES - Portal Application

- **FAR Details (Proposed):** Specifies the proposed floor area ratio details.
 - **Permissible:** Enter the permissible FAR.
 - **Premium:** Enter the premium FAR.
 - **TDR:** Enter proposed Transfer of Development Rights details.
- **Site Condition:** Specifies the existing site condition.
 - **Water:** Here provide the source of water.
 - **Well:** Choose this option, if the source of water is well.
 - **BWSSB:** Choose this option, if the source of water is Bangalore Water Supply and Sewerage Board.
 - **Power:** Here provide the source of power.
 - **KPTCL:** Choose this option, if the source of power is Karnataka Power Transmission Corporation Limited.
 - **Electricity HT/LT:** Select whether electricity facility available or not.
 - **Future Connected Analytics (FCA) related active deployment** of Planned Road signs/Relatedly Live Screens/Billboards/ Hoardings/Signages, where these views or insights can guide any FESA service or SAR experience

NOC Blocks of Buildings with Strategic-Tactical-Operational intelligence Infrastructure and/or Data Management influencers

Perspective Imagery (PI) Indicators for a road system/route

- PIEV - Perception time, Intellection time, Emotion time, Volition (Final action) time
- Perception time - time required to perceive a situation or object
- Intellection time - time required to compare different thoughts, regroup thoughts and different points of understanding, register new "information, thoughts or sensations"
- Emotion time - time required to compare "emotional responses, sensations or disturbances"
- Volition time - time required for final action
- Proposed **Traffic Engineering (TE) Designed Synergy time – Viewpoint time required to evaluate issues or incidence resolution / management for a RADIUS OF COVERAGE**
- PIEV* time required depends upon aspects such as
 1. Physical characteristics of the driver
 2. Psychological factors influencing or affecting the driver, savings & safety interests
 3. Environmental conditions, influencers, situations, road & traffic health
 4. Purpose of trip, trip planning,
 5. Type and speed of vehicle, condition and adherence to norms
 6. New NSSR Theme for Road Safety and choices for traffic health
 7. (Occupation based or Trends based) Self-assessment for fitness, drive guidance
 8. Availability of feedback systems 9. Editing of RADIUS of coverage for TE Viewpoint Synergy



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Relative understanding of KSFES FOR NSSR-RS / RI-FESA NOC Culture

-



Items and sub-items code for Time of the Year	Items and sub-items for Time of the Year
11	Weather conditions
12.1 to xyz	Climate change dynamics
18.1 to xyz	Hazards level tabulations
Items and sub-items code for Risk Elevation	Items and sub-items for Risk Elevation
21	Specific S&A Risk
22	EOCC Connected S&A Risk
28	Hour-glass V2R Effectiveness Risk